Fiscal Year:	FY 2016	**Task Last Updated:**	FY 09/14/2015
PI Name:	Ferl, Robert Ph.D.		
Project Title:	Early Stage Plant Adaptation to Spaceflight - Molecular Responses of Arabidopsis to the Transition from Terrestrial Environment to Space		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Paul, Anna-Lisa Ph.D. (University of Florida)		
Grant/Contract No.:	NNX15AB12G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The objective of this proposal is to use suborbital spaceflights in SpaceShipTwo to develop an understanding of the genomic and metabolic plant responses upon entry into spaceflight. The long term goal of this line of research is a comprehensive appreciation of the biological responses that occur as terrestrial life experiences phases and aspects of spaceflight. We will build on our previous genomic work using Arabidopsis plants by imaging green fluorescent protein (GFP)-tagged promoters from specific genes indicated as potentially involved in early spaceflight transitions and by fixing plants during suborbital flight for whole genome expression analysis with arrays and RNASeq. This project makes extensive use of a unique middeck locker imager system that was developed within the NASA Flight Opportunities Program, an imager that is capable of delivering multispectral fluorescent images at high resolution within space vehicles. The project also utilizes the specialized experience of the Principal Investigator (PI) team in deploying plant		

	molecular biology gene expression experiments and imaging in diverse space-related environments. These studies are innovative in that they will provide the first molecular genetic data from suborbital spaceflights, thereby filling a substantial gap in knowledge regarding the early stages and temporal dynamics of gene expression responses to spaceflight. Specifically these data will provide insights into the effects of early stage spaceflight responses so as to begin to round out a more comprehensive understanding of the temporal and transitional plant responses to spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	To gain fundamental insight into how plants perceive and process the signals that control tropism and adaptive physiology in novel and extreme environments. It also provides a unique understanding of how a terrestrial organism transitions from a normal gravity environment to one outside its evolutionary experience. Currently only suborbital vehicles can facilitate access to this unique window of transition for data collection. This research also highlights how suborbital vehicles can be used to optimize experimental design to help maximize International Space Station (ISS) utilization in future experiments.
Task Progress:	Background and summary of parabolic flights The past year has provided many different avenues to expand our knowledge of both the equipment needed for sub-orbital flight and the understanding of plant responses to early flight. The experiment was awarded with a parabolic flight campaign (53-SP Gene Expression) to test the hardware and biology expectations of future sub-orbital campaigns. The overall goal in the campaign was to investigate alternative development pathways for the FLEX imager prototype and validate the science applications for those pathways. Successful testing of the FLEX hardware within the parabolic flight environment immediately raises the technology readiness level of the system to 7 or 8. After this campaign, future parabolic flight experiments could then make essentially direct use of the FLEX imager and the floor-mounted base-plate and setup used in this campaign for future science campaigns of biological significance. This campaign allowed the hardware to first be vetted in the parabolic platform and shown to be sensitive enough to detect meaningful biological events that happen in the parabolic flight timeframes and environments. There were two associated objectives of the parabolic campaign: First, to further develop the genome level, genetic analyses initiated with earlier parabolic flight and orbital experiments. Numerous genes associated with Auxin metabolism are up-regulated in response to both parabolic flight and spaceflight. The genetic analyses in the parabolic flight will contribute to the further dissection of this response. During the flights, plants engineered with gene reporters designed to respond to variations in auxin metabolism were imaged with the FLEX imager. Additional plants were preserved to confirm telemetric data obtained with the FLEX imager. Second, we hoped to obtain a replicate data-set characterizing the extent of thermal variation on the surface of Arabidopsis leaves during parabolic flight. Previous flights were very successful in this endeavor; our experiments demonstrated that if air is not allowed to circulate that the surface leaf temperature can vary by 2-4°C between the 0g and 2g segments of the parabola. It was our goal to repeat these experiments using the next generation FLIR (forward looking infrared) thermal imaging devices in conjunction with plants engineered with gene reporters designed to reveal changes in gene expression in response to heat-induced stress. Data was collected in the form of digital images and tissue-specific, reporter gene expression patterns and all will serve to calibrate the primary function of the FLEX imager. These experiments utilized Arabidopsis engineered with GFP (green fluorescent protein) gene reporters so real-time images can be collected telemetrically (monitoring GFP gene expression). FLEX Imaging High quality FLEX images (GFP) were recovered during flight. Flight data tracked using the MSR program was collected every day. Plants for FLEX image verification were harvested at each turn during the flight and will be used for image corroboration. FLEX images were captured every 60 seconds during flight. The excitation lights were turned on approximately 3 seconds prior to image capture and then promptly turned off until the next image was taken to keep the imager unit cool, and the lights from heating. Images were recorded for the entire flight all four days starting before take-off. Testing of the newly implemented software meant that the recording program ended on different days at different times, but all parabolas were captured and recorded before the program was stopped or paused. Software Development FLEX technology development was considerably advanced during the past year due to software development for the deployment of the hardware on future suborbital flights. The RSA software, used to control the different systems for each of the units in the FLEX, was developed and subsequent parabolic flight allowed for refinement of the program. The software will allow for crew autonomous use of the hardware in flight. The software GUI allows simple push button stop and start of the imager, as well as the ability to manipulate frequency and type of data captured. Software worked well during the flight, and with a few small modifications will be ready for sub-orbital flight in the coming months. This system will also be fully automated to be used in other experimental situations. Gene Expression during Early Flight In preparation for future sub-orbital flights work is being completed to fully understand molecular response in early flight. In order to study early reactions to flight, plants were loaded in to KFT's prior to take-off and were fixed in rapid succession during the first maneuvers of parabolic flight in an attempt to capture gene expression patterns induced by early responses to gravitational changes RNA extracted from these plants has been extracted and analyzed using microarrays to determine gene expression levels. These arrays will be analyzed over the coming months. Genes that are identified to be quick responding genes will be used for possible future production of transgenic plants to facilitate understanding of the genes localization, pathways, and the roles they play in a plants response to early flight. Thermal Variation The FLEX imaging system is also equipped with FLIR (Forward Looking InFrared) technology that allows visualization of thermal variation of plants (or a subjected located in the imager). Past research has seen that microgravity tends to cause a thermal increase in leaves of plants. This is believed to be due to lack of air circulation over the leaves during zero-gravity. During this flight campaign a 14-3-3 protein deficient mutant line was used in addition to the wild-type Arabidopsis lines to help better understand the significance of the results. Analysis of these results is on-going.

	Conclusions and future work The experiment and preparations are ongoing, proceeding smoothly and on schedule. Over the coming year the project will continue analysis of microarray data from the parabolic campaign of 2015. Development of transgenic plants with GFP biomarkers for genes identified as responsive to early flight will be prepared. Finalizing the software development for flight will be completed in the next year.
Bibliography Type:	Description: (Last Updated: 11/27/2022)
Articles in Peer-reviewed Journals	Ferl RJ. "Suborbital vehicles to study transition adaptation to spaceflight - Why biologists should care about the new suborbital flight opportunities." Gravit Space Res. 2014 Dec;2(2):58-65. http://gravitationalandspacebiology.org/index.php/journal/article/view/658 , Dec-2014
Articles in Peer-reviewed Journals	Bamsey MT, Paul A-L, Graham T, Ferl RJ. "Flexible imaging payload for real-time fluorescent biological imaging in parabolic, suborbital and space analog environments." Life Sci Space Res. 2014 Oct;3:32-44. http://dx.doi.org/10.1016/j.lssr.2014.09.002 , Oct-2014